

Supaporn Klabklaydee

DOCTORAL STUDENT · AI/ML, HPC & SCIENTIFIC SOFTWARE

1-1-24, Minami-ku, Yokohama, Japan

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“Scalable AI/ML & HPC — software you can run, test, and reproduce”

Summary

PhD researcher (Institute of Science Tokyo) building **scalable AI/ML systems, HPC workflows, and tested reproducible software**: Git/GitHub, Docker/Singularity, Snakemake, SLURM multi-node GPU (A100/H100/V100), Ray, FAISS. Open profile: github.com/Suppboat2544 (**4** public repos) — flagship GSAT toxicity GNN (published eval suite; 1 star, 1 fork). DOE Tier-1 HPC (LLNL DSSI) + national HPC collaboration (RIKEN): distributed training/inference, containerised DAGs, Git-LFS datasets, evaluation harnesses. Cheminformatics / environmental molecular systems are application domains; Computing artifacts below are primary.

Education

Institute of Science Tokyo

Tokyo, Japan

PH.D. CIVIL & ENVIRONMENTAL ENG. — **AI/ML & CHEMOINFORMATICS CONCENTRATION**

Sep. 2025 – Present

- GPA: **4.05 / 4.5**. Dissertation: CRAM Digital Twin — property-conditioned VAE+GAT+GRU-SELFIES; Graph Transformer biotransformation (210K+ pairs, 87% Top-3); kinetic PINNs; advised by Prof. M. Fujii.
- **AISpread1000** doctoral researcher (¥5M, PI: Fujii): million-scale diverse-DOM generation via k -NN chemical-space learning; open/reproducible generative ML pipelines.

Institute of Science Tokyo

Tokyo, Japan

DATA SCIENCE & AI — EXPERT LEVEL PLUS (UNIVERSITY-WIDE)

2025

- Highest certification tier: supervised/unsupervised ML, deep learning, statistical modelling, AI applications in science & engineering.

Institute of Science Tokyo

Tokyo, Japan

M.ENG. CIVIL & ENVIRONMENTAL ENGINEERING

Oct. 2023 – Sep. 2025

- GPA: **4.16 / 4.5**. Thesis GSAT GNN → ACS ES&T Water ($R^2=0.907$; 98.9% faster; 4.7M reaction steps / 12 pollutants).

KMUTT

Bangkok, Thailand

B.ENG. ENVIRONMENTAL ENG. (INT'L), 1ST CLASS HONOURS, DEAN'S LIST

2019 – 2023

- GPA: 3.75 / 4.0. Best Rising Star (Physical Sciences). Thesis: molecularly imprinted TiO_2 + DFT (Gaussian) + RF/SHAP rate prediction (RMSE 0.042 min^{-1}).

Selected Computing Experience

JAMSTEC & RIKEN

Remote / Japan

REMOTE RESEARCH COLLABORATOR — HPC BIOINFORMATICS & DISTRIBUTED ML

Dec. 2025 – Present

- Ocean-scale corpora (TARA, OSD, GOS, Malaspina); dual-embedding: ESM-2 (480-d, FP16) + k -mer ($k=4$, 8K-d), Mahalanobis fusion on **1,293,763 sequences**; chunked batching + HDF5 mmap I/O.
- HPC clustering: HDBSCAN + **FAISS IVF-PQ** GPU ANN ($O(n \log n)$); <6 h wall time; **847** families (78.3% EC consistency; 156 novel; Silhouette 0.47 vs. 0.32); UMAP stratified by depth/temp.
- XGBoost+SHAP on cluster physicochemical summaries — temperature–depth explains **34.7%** inter-family embedding variance.
- RIKEN software practice: **Singularity**, **Snakemake** DAGs, Git-LFS, SLURM job arrays; shared reproducible enzyme-atlas workflows.

JST × LLNL Data Science Summer Institute

Livermore, CA, USA

HPC / AI RESEARCH INTERN (JST EXCHANGE)

Jul.–Sep. 2025

- DOE Tier-1 (El Capitan / Lassen): multi-node **A100/H100** via SLURM; structure-conditioned GA (SMILES/SELFIES graphs) + ESM-2 fitness + AlphaFold2 pLDDT proxy for antigen design (w/ Conti, Faissol, da Silva).
- Distributed PyTorch / **torch.multiprocessing** / **Ray** on 8× A100 — 4.2 h → 18 min/gen (**14×**); NSGA-II Pareto + GP active learning (10,000 → 47 candidates).
- XAI: Integrated Gradients + BERTViz attention; shipped **Snakemake + Singularity** replay (FASTA → ranked library; LLNL GitLab). **Best Student Cohort** / Top Presenter.

- **CRAM generative stack:** VAE + GAT encoder + GRU-SELFIES decoder; median isomers/formula 1.4 → 9.3; Graph Transformer pathways: 210K+ pairs, 99 classes, Focal Loss + SMILES aug. + SWA.
- **PINN** ocean-depth kinetics ($R^2=0.93$; 0–6000 m; 17,500× half-life span); validated vs. ARGO/GLODAP (300K+). Hybrid GAT+MPNN half-life ($R^2=0.847$) + Gradio demo.
- **MOSAIC-GNN** (in prep.): heterogeneous MS1 formula assignment (mass diffs, isotopes, elemental constraints). Quantum fidelity-kernel null study (PennyLane 8-qubit vs. RF/RBF-SVM/AttentiveFP; 280 conditions).

Faculty of Medicine, Chulalongkorn University (NRCT)

Bangkok / Remote

COMPUTATIONAL RESEARCH ASSISTANT — GNN / DOCKING SOFTWARE

Mar. 2022 – Present

- 2,052 SMINA/Vinardo dockings (9 ligands × 76 sites × 3 pH) + 120 MMP-1 runs; GROMACS MD (CHARMM36, 100 ns); PyG heterogeneous GNN multi-task CSI (RMSE 0.55 kcal/mol, $\rho=0.95$).
- Integrated Gradients pharmacophore recovery; open repo Collagen-Gallic-Acid-Crosslinking.

IST • JAXA • JAMSTEC • University of Liverpool

JP / UK

WORKSHOP INSTRUCTOR & INVITED LECTURER — SOFTWARE / ALGORITHMS

2025 – 2026

- **12 classes:** IST RDKit+GNN (2); JAXA graph theory for satellite/stellar networks — shortest path, centrality, Laplacians, NetworkX (3); JAMSTEC metagenomics/ESM-2/HDBSCAN (3); Liverpool invited cheminfo+bioinfo notebooks (4).
- Teaching code: RDKit_Workshop (SMILES/SELFIES, ECFP, Tanimoto, UMAP, MoleculeNet-style GNN baseline).

Software & Open Source

Profile: github.com/Suppboat2544 — 4 public repositories.**GSAT** — GSATmodel_Evaluation_ToxicityLC50_emergingpollutants: published GNN LC50 model + evaluation/tests (1 star, 1 fork).**RDKit Workshop** — RDkit_Workshop: instructional molecular software pipelines.**Collagen GNN** — Collagen-Gallic-Acid-Crosslinking: docking + GNN selectivity.**Site / portfolio:** Supapornboat.github.io. Practices: Git-first, Docker/Singularity replay, Snakemake, SLURM arrays, Gradio, AWS/GCP spot, CI-friendly eval harnesses.

Selected Publications & Awards

S. Klabklaydee et al., “Multigenerational Toxicity Prediction Using Graph Neural Networks...”, *ACS ES&T Water*, Jan. 2026.Co-author: B. Zhang et al., “Accurate Detection...Unknown PFAS Transformation Products...”, *Water Research*, 2025.S. Klabklaydee et al., “Divergent CRAM Formation and Persistence Across Freshwater and Marine Environments Revealed by AI-Expanded MS/MS Libraries,” *ChemRxiv*, 2025. doi:[10.26434/chemrxiv.15007942/v1](https://doi.org/10.26434/chemrxiv.15007942/v1) [Full text]S. Klabklaydee, N. Sudsa-Ard, M. Fujii, carboxyl-metabolizing enzymes (active site + adaptation), *ChemRxiv*, 2025.

In prep. (1st): MOSAIC-GNN; CRAM Digital Twin / kinetic PINN; marine enzyme dual-embedding (JAMSTEC/RIKEN); collagen GNN docking.

Awards: Best Student Cohort, JST×LLNL DSSI 2025; MEXT Full Scholarship (Master–PhD); AISpread1000 (¥5M); Expert Level Plus (DS&AI); Intel ISEF 2019 (Thailand); I-Envex Gold 2018.

Skills (Computing-first)

Programming	Python, R, Julia, C++, C, Java, Bash, Git / GitHub
ML / DL	PyTorch, PyG, Hugging Face (ESM-2), Scikit-learn, GNN/GAT/MPNN, VAE, PINN, Graph Transformer, XGBoost, Ray
HPC / Dist.	SLURM, multi-node GPU (A100/H100/V100), MPI, FAISS IVF-PQ, HDF5 mmap, job arrays
Software Eng.	Docker, Singularity, Snakemake, Git-LFS, reproducible pipelines, testing & evaluation suites
Cloud / UI / XAI	AWS / GCP (spot), Gradio, SHAP, Integrated Gradients, Captum, BERTViz
Data / Viz	UMAP, DBSCAN/HDBSCAN, Pandas, NumPy, Matplotlib, NetworkX
Cheminfo / Sim.	RDKit, DeepChem, SELFIES, OpenBabel, AutoDock Vina, GROMACS, Gaussian (DFT), PennyLane
Office	Microsoft Word, Excel, PowerPoint
Domains	Machine Learning, HPC for Science, Chemoinformatics, Environmental AI
Languages	Thai (native), English (professional), Japanese (intermediate)